
Vortex merger: vorticity | t = 0.0000 | frame 1/61
Re = 350 | vortex separation = 0.90 | 128 x 128 evaluation grid | identical color scale across all panels
GT / DNS
Gauge-canonical Eigenfluid
Eigenfluid without gauge fixing
Analytic Laplacian
64 modes
Reference DNS
Velocity MSE 2.804e-05 | Vorticity MSE 1.819e-03
Velocity MSE 2.954e-05 | Vorticity MSE 2.044e-03
Velocity MSE 2.541e-05 | Vorticity MSE 1.595e-03
GT / DNS
Gauge-canonical Eigenfluid
Eigenfluid without gauge fixing
Analytic Laplacian
1024 modes
Reference DNS
Velocity MSE 1.846e-07 | Vorticity MSE 1.333e-04
Velocity MSE 3.199e-07 | Vorticity MSE 2.335e-04
Velocity MSE 1.385e-19 | Vorticity MSE 9.396e-17
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Same GT frame 0 projected into each basis; autonomous Navier-Stokes Galerkin rollout thereafter. No later GT refitting.